Prestudy - Calibrating a Helmholtz coil and function tests of cryogenic PCB

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Bachelor's thesis project -

A cryogenic magnetic field sensor for quantum computing

1 Introduction

This prestudy presents the background theory and methodology for calibrating a constructed Helmholtz coil. Furthermore, the coil will be used to calibrate the hall - element for later use with the entire magnetic field sensor system.

1.1 The Helmholtz coil

A Helmholtz coil is a pair of coils with a certaing geometry which makes it ideal for creating a homogenous magnetic field with a highly controllable magnetic flux density. A schematic picture of the coil is shown in figure 1.

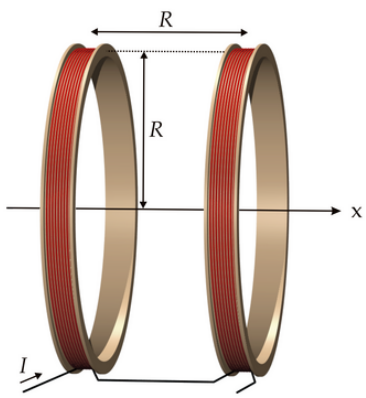

Figure 1: *Schematic figure for a helmholtz coil. The setup consists of two coils of equal geometry. By placing the coils the same distance apart as their individual geometry, a homogenous magnetic field is created in the center. By running the same current through both coils, the magnetic flux density can be easily computed.*

By placing the coil with a distence apart that is the same as their individual radius, the magnetic flux density in the center of the coils can be calculated as:

$$B = \left(\frac{4}{5}\right)^{3/2} \frac{\mu_0 N I}{R}, \tag{1}$$

with μ_0 being the magnetic permeability, N the number of turns per coil, I the current and R the radius.

The group has already counstructed a helmholtz coil. For this coil, the radius is ≈ 3 cm and each coil has aroun 100 turns (we might have lost count at some point during the winding). The coil has been verified to conduct current and has a resistance of around 2 Ω each. Using the geometry of our coil and expressing the magnetic flux density B as a function of the current I, the expression would be as follows:

$$\left(\frac{4}{5}\right)^{3/2} \frac{\mu_0 N}{R} = 2.6 \cdot 10^{-3} \rightarrow B = 2.6\, I \cdot 10^{-3} \tag{2}$$

We should therefore expect to see approximately 2.6 mT/A in the center of our coil.

1.2 The Hall element

To measure the magnetic field, a commercial Hall effect sensor is used. The sensor is the +++ from AKM. It has been prepared for calibration by being mounted to a small pcb, as shown in figure

By providing a current through the hall element, a measureable Hall voltage should be produced. The formuala for the hall-voltage is given as

$$V_H = \frac{BI}{dnq} \tag{3}$$

with the magnetic flux denisty being B, the current I. d is the thickness of the hall element, q is the charge of the charge carrier and n is the charge carrier density. Since d, q and n all are constant (n changes as a function of temperature, and we can assume the temperature is constant), the hall voltage should be linearly proportional to both the magnetic flux density and the current.

Thus, by providing a steady current to our hall element we should be able to calibrate the hall voltage to the changing magnetic flux density, which we can control through our Helmholtz coil.

2 Methodology

3 Expected results